Literature Review

Marine debris is a serious problem that affects marine life and underwater environments. Autonomous underwater vehicles (AUVs) can help detect and remove rubbish from the ocean. These vehicles use cameras and deep learning models to identify underwater objects. However, detecting rubbish underwater is difficult because of poor lighting, blurry images and particles floating in the water. Marine plants and animals can also look similar to rubbish. This makes it harder for detection models to identify objects correctly. Researchers have developed different deep learning methods to improve underwater debris detection.

Jiang et al. [1] studied how to improve underwater debris detection using YOLOv8. They used the TrashCan dataset, which contains 7,212 annotated underwater images collected from JAMSTEC's deep-sea video database. The images include rubbish, marine animals, plants and underwater robot parts. The dataset has two versions. TrashCan-Instance classifies rubbish by object type, such as bottles and cans. TrashCan-Material classifies rubbish by material, such as plastic, metal and rubber. The authors made three main changes to the original YOLOv8 model. First, they added a CloFormer module to help the model identify small details and use information from surrounding areas. Second, they used a C2f-SCConv module to reduce unnecessary calculations and improve feature extraction. Third, they replaced the original bounding box loss function with WIoU v3 to improve object localisation. The improved model achieved a 5.7 percentage-point increase in mAP@0.5 and a 6 percentage-point increase in F1 score on TrashCan-Instance. It also achieved improvements of 5.5 and 5 percentage points on TrashCan-Material. These results show that the changes improved detection accuracy. However, the model still struggles with small or partly hidden objects. The extra modules also make the model more complex. The authors mainly tested the model on TrashCan, so its performance in other underwater environments is still unclear.

While [1] focused on improving the YOLOv8 model, Đuraš et al. [2] focused on creating a more diverse dataset for underwater debris detection. They introduced the SeaClear Marine Debris Dataset, which contains 8,610 underwater images collected from five locations in Croatia and France. The images were captured using remotely operated underwater vehicles (ROVs). Unlike TrashCan, which mainly contains deep-sea images, SeaClear focuses on shallow-water environments. The images include different lighting conditions, water clarity and backgrounds. The dataset originally contains 40 object categories, including rubbish, marine animals, plants and underwater robot parts. It also provides bounding box and segmentation annotations. For their detection experiments, the authors combined categories with fewer than 50 annotations. This reduced the number of categories to 34. They trained and tested two object detection models, Faster R-CNN and YOLOv6. Both models were pretrained on the COCO

dataset before being trained further on SeaClear. Faster R-CNN detects objects in two stages, while YOLOv6 uses a single stage and is generally faster. The results showed that YOLOv6 achieved approximately 7 percentage points higher detection performance than Faster R-CNN on the standard dataset split. The authors also tested underwater image enhancement by adjusting colour and contrast. However, clearer images did not always improve detection accuracy. They also found that model performance dropped when images came from different cameras or locations. This shows that detecting rubbish in unfamiliar underwater environments is still challenging.

Both studies [1] and [2] show how deep learning can help detect marine debris. However, they focus on different problems. Jiang et al. [1] improved YOLOv8 to increase detection accuracy using the TrashCan dataset. In comparison, Đuraš et al. [2] created the SeaClear dataset and tested how well detection models work in different underwater conditions. The first study shows that changing a model's architecture can improve accuracy. However, it does not fully show how well the model works in other underwater environments. The second study shows that dataset diversity is important. It also highlights problems such as class imbalance and lower accuracy in unfamiliar conditions. Both papers are useful for our project because we are using the TrashCan dataset to detect and classify marine debris. Our project focuses on comparing Faster R-CNN and RetinaNet instead of improving YOLOv8. We also use underwater image augmentation to help the models handle different image conditions. Repeat-factor sampling is used to reduce the effect of class imbalance during training. These studies help us understand the challenges of underwater debris detection and the methods that may improve model performance.

Bibliography

[1] Jiang et al., "Research on the Identification and Classification of Marine Debris Based on Improved YOLOv8," 2024.

[2] Đuraš et al., "A Dataset for Detection and Segmentation of Underwater Marine Debris in Shallow Waters," 2024.